



Course Title: STATISTICS and PROBABILITY

Course Code: ELEC CS-FE2

Course Credit: 3 units

Pre-Requisite: MMW, Basic Calculus

Course Description:

This course provides fundamental concepts of statistics and probability with applications in computer science. Topics include descriptive statistics, probability theory, probability distributions, inferential statistics, and their applications in data science, machine learning, and algorithm analysis.

Course Learning Plan

Week	Topics and Subtopics
Week 1	Orientation
Week 2 - 4	Introduction to Statistics and Data Analysis - Definition and importance of statistics in computing - Dataset - <i>Exploratory Data Analysis</i>: - Univariate and Bivariate Analysis - Numerical Measures - Visualization - Basics of R Software - Group Assessment
Week 5 - 6	Basics of Probability Theory - Importance of Probability to Inferential Statistics - Sample Space, Events, Operations on Events - Counting principles: Permutations and Combinations - Bayes' Theorem and conditional probability
Week 7	Quiz 1
Week 8 - 9	- Random variables and expectation - Mathematical Expectation - Chebyshev's Theorem
Week 10 - 11	Probability Distributions - Discrete probability distributions: Binomial - Continuous probability distributions: Normal
Week 12	Quiz 2





Week	Topics and Subtopics
Week 13 - 14	Inferential Statistics - Hypothesis testing - Parametric and Non Parametric Tests
Week 15	Quiz 3
Week 16	Final Exam

Reference:

- Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2012). Probability and statistics for engineers and scientists (9th ed.). Boston, MA: Pearson.

Prepared by:

Perlyn Mae R. Dilla
Faculty Member
Department of Mathematics and Statistics

Noted by:

Kenneth James T. Nuguid
Chairperson
Department of Mathematics and Statistics

Approved by:

Dr. Lourdes Alvarez
Dean, College of Science

